

Lab Report 2

Modelling and Identification of a DC Motor

Robotics Lab: Control Systems

IRO-4 SS26

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The DC motor model can be represented using electrical and mechanical equations, which are implemented in a Simulink block diagram. The aim of this experiment is to determine the unknown model parameters from measured data and validate the model through simulation.

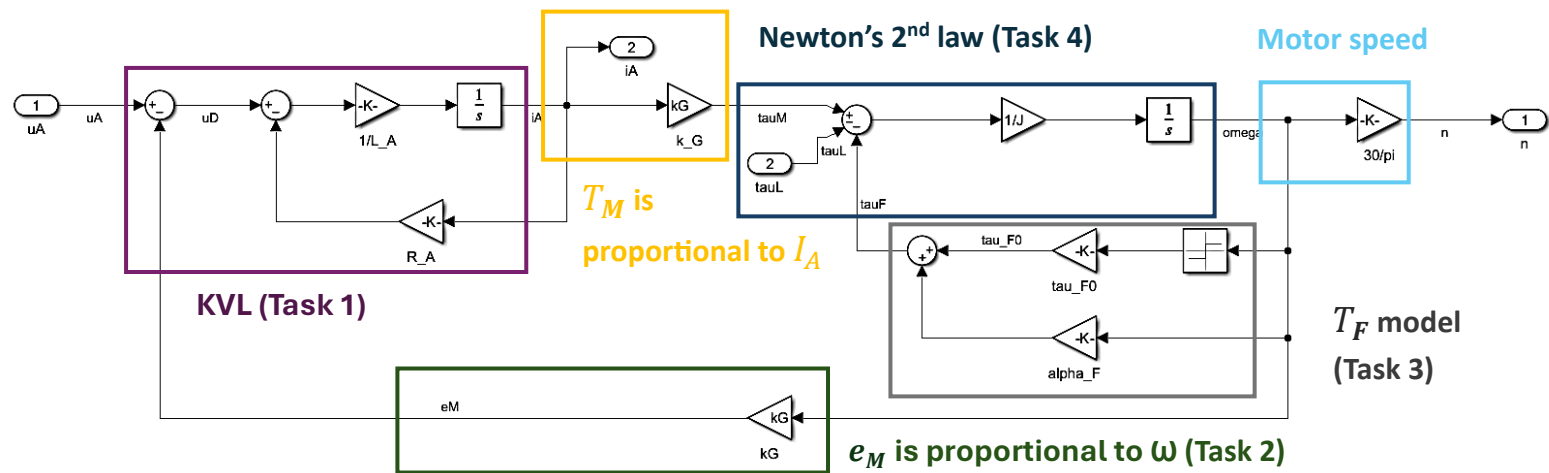


Figure 1: Block diagram representation of derived model

The experiment is carried out using a laboratory DC motor setup, where armature voltage is applied and current and speed are measured.

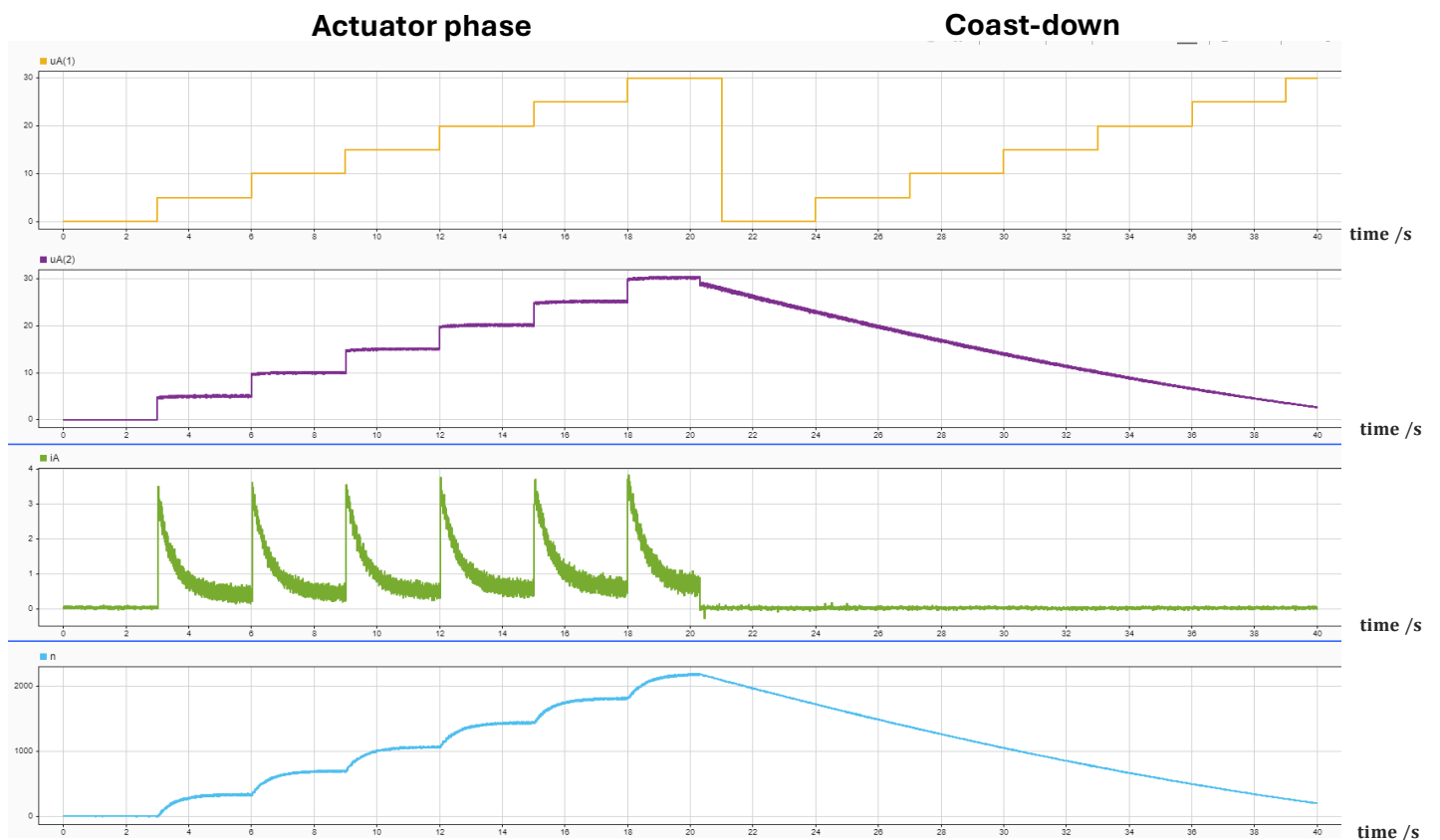


Figure 2: Measured armature voltages, armature current and rotational speed

Based on these measurements, parameters such as resistance, inductance, friction coefficients, motor constant, and inertia are identified.

Task 1: Determination of Armature Resistance R_A and Inductance L_A

Using Kirchhoff's Voltage Law, the armature circuit is described as a first-order (PT1) system when the rotor speed is approximately zero:

$$\frac{I_A(s)}{U_A(s)} = \frac{1}{R_A} \cdot \frac{1}{\frac{L_A}{R_A}s + 1}$$

Thus:

$$\text{Steady-state gain: } K = \frac{1}{R_A} = \frac{i_{\infty}}{u_{\infty}}$$

$$\text{Time constant: } T = \frac{L_A}{R_A} = 0.048 \text{ s}$$

Results

- $R_A = 1.4276 \Omega$
- $L_A = 0.068 \text{ H}$

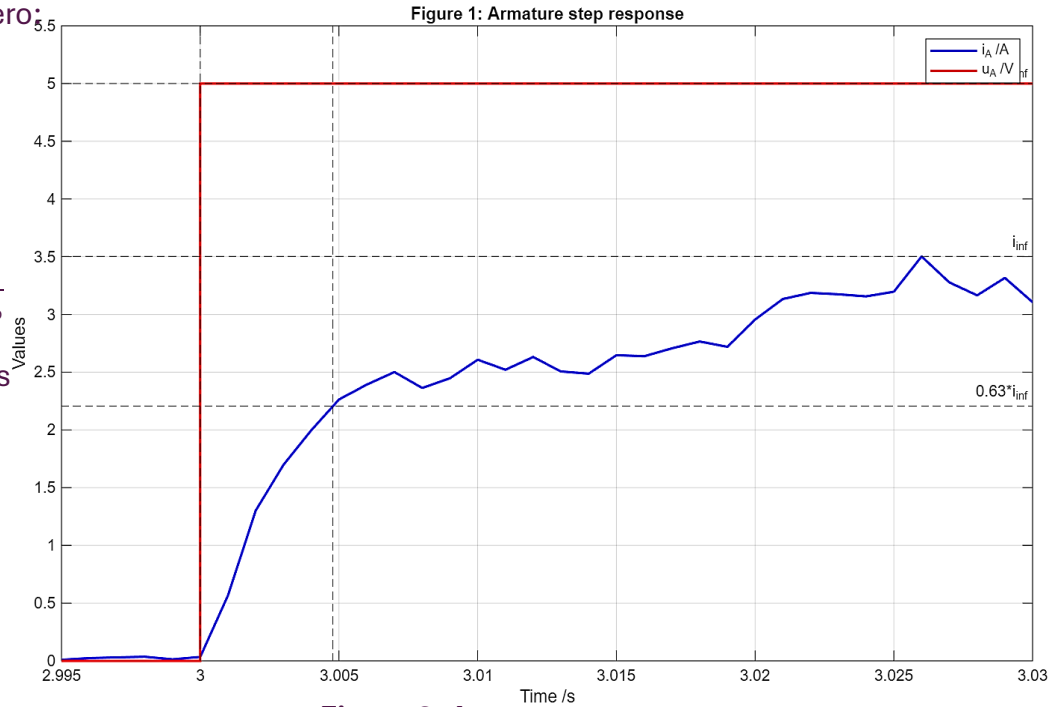


Figure 3: Armature step response

Task 2: Determination of Motor/Generator Constant k_G

The motor constant k_G is defined by the proportional relationship between induced emf and angular velocity: $e_M = k_G \cdot \omega$. During the **coast-down phase** (the motor is disconnected from the power amplifier after ~ 20 s), the armature current is zero ($i_A = 0$), so the terminal voltage equals the induced emf. Therefore, k_G can be determined from the slope of the linear relationship between e_M and ω . To reduce noise effects, e_M is plotted against ω , and a linear fit is applied.

Result $k_G = 0.1268 \text{ Vs/rad}$

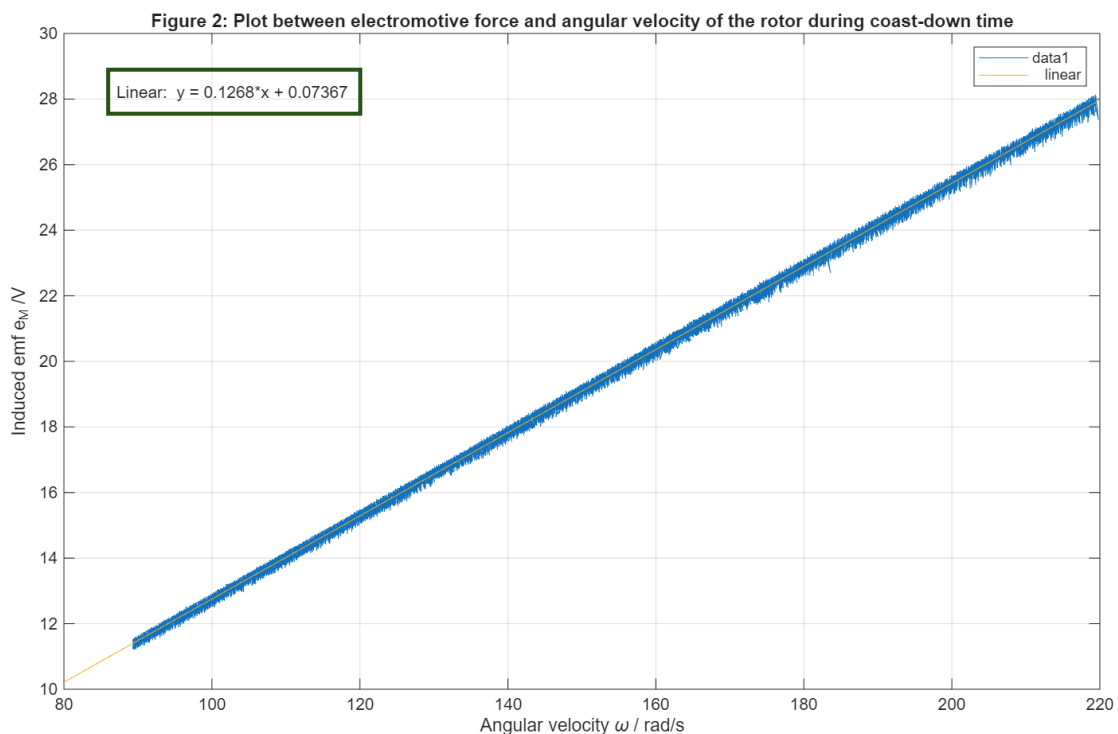
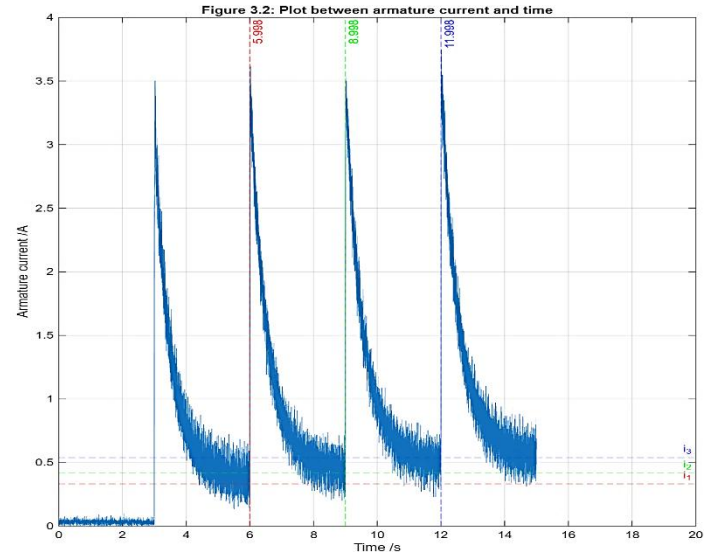
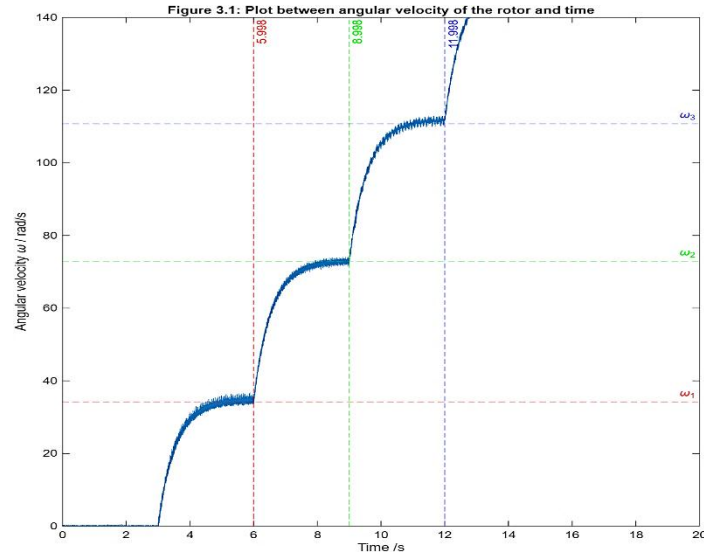


Figure 4: Plot between electromotive force and angular velocity during coast-down phase

Task 3: Determination of Dry Friction T_{F0} and Coefficient of Viscous Friction α_F

Under **no external load conditions** ($\tau_L = 0$) and steady-state operation ($\frac{d\omega}{dt} = 0$), the motor torque balances the friction torque: $\tau_F = \tau_M = k_G \cdot i_A$. Thus, the friction torque can be computed from measured armature current using the motor constant k_G . To reduce noise, steady-state values are extracted at the end of each voltage step. Three operating points are selected after transients have settled.

Figure 5: Plots between angular velocity, armature current and time



Although averaging over a short interval could improve accuracy, care is taken to avoid intervals where external transient effects are still present.

The friction model is assumed linear: $\tau_F = \tau_{F0} + \alpha_F \omega$

A linear fit is applied to the data points to determine the parameters.

Results

Experimental data points

ω (rad/s)	τ_F (Nm)
34.2213	0.0422
72.9064	0.0528
110.734	0.0683

Dry friction torque: $\tau_{F0} = 0.02963$ Nm, Viscous friction coefficient: $\alpha_F = 0.0003418$ Nm/(rad/s)

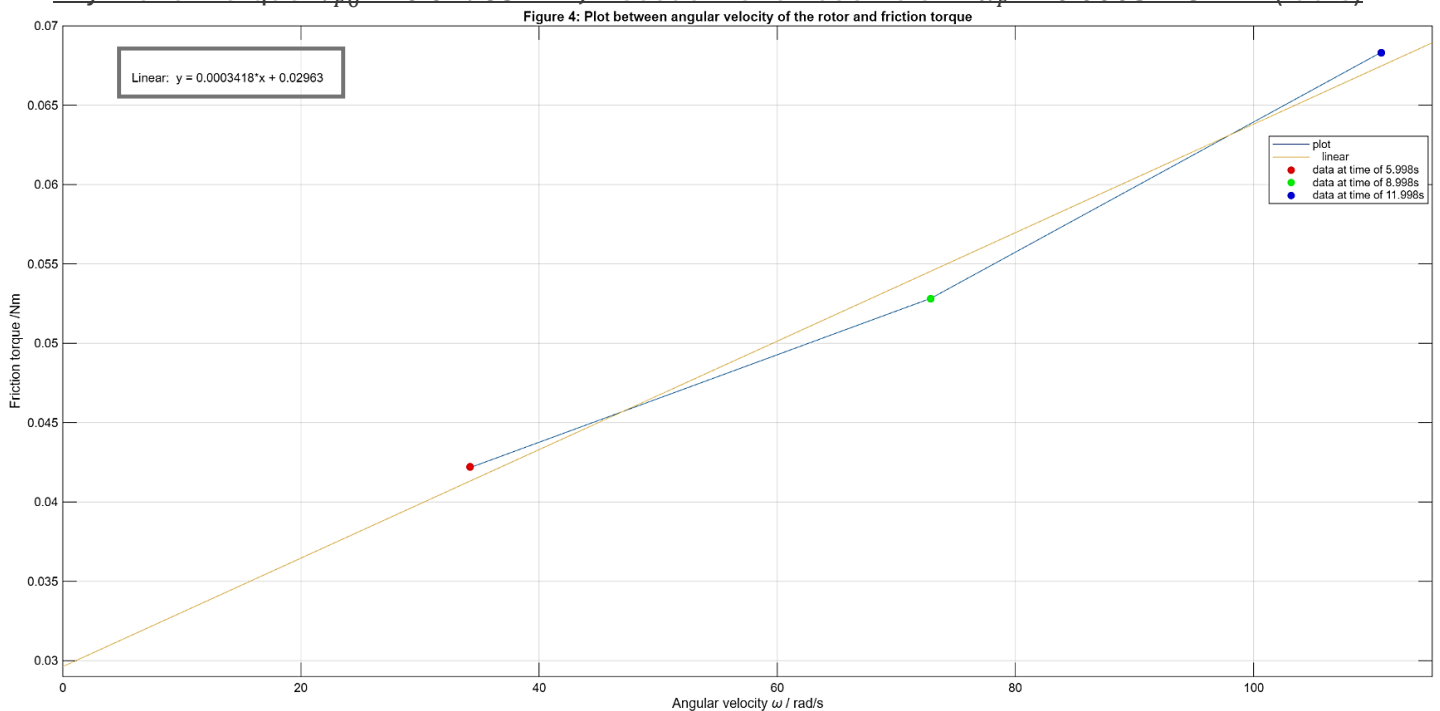


Figure 6: Plot between angular velocity and friction torque

Task 4: Determination of Moment of Inertia J

During the **coast-down phase**, the armature current is zero ($i_A = 0$), hence the motor torque becomes zero: $\tau_M = 0$. Angular velocity $\omega(t)$ is plotted against time and a tangent is drawn at a selected operating point ω_p . Angular acceleration is determined: $\frac{d\omega}{dt} = -11.86 \text{ rad/s}^2$ and friction torque at ω_p is computed using: $\tau_F(\omega_p) = \tau_{F0} + \alpha_F \omega_p$.

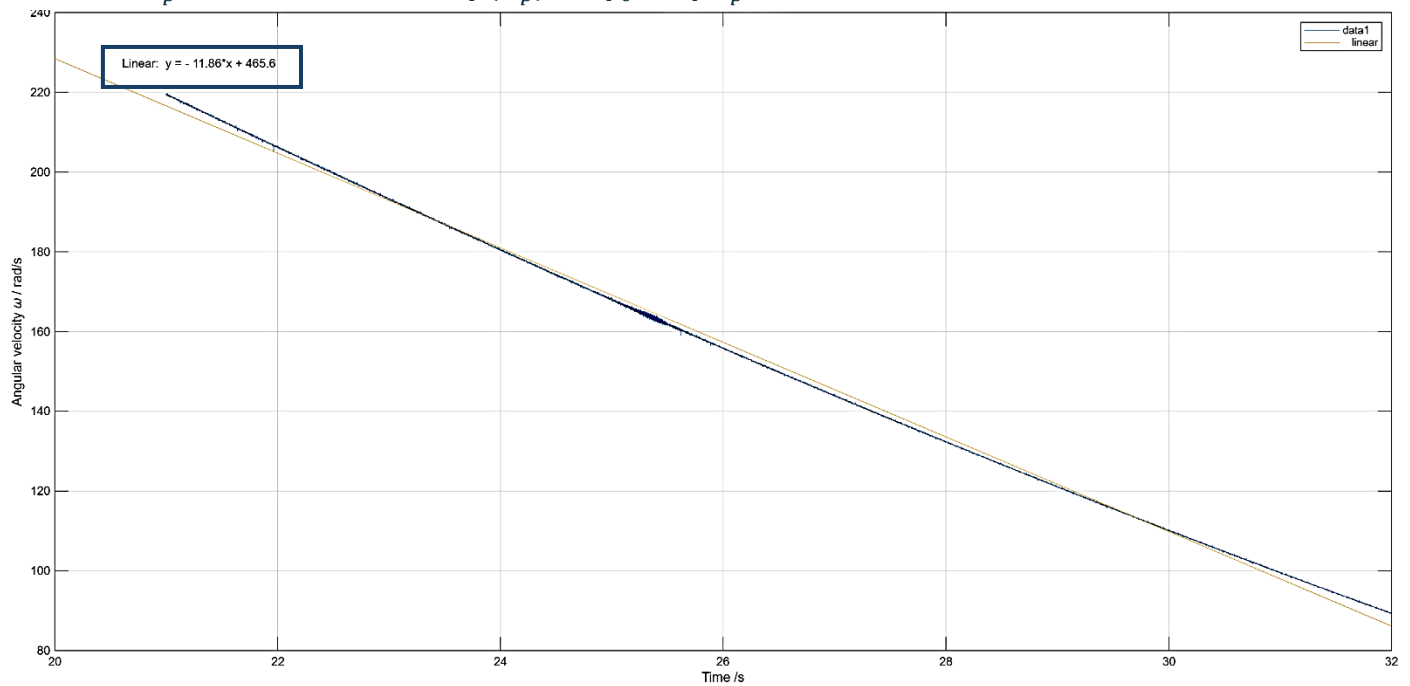


Figure 7: Plot between time and angular velocity

The rotational dynamics are then governed only by friction: $J \frac{d\omega}{dt} = -\tau_F$. Thus, the moment of inertia can be calculated as: $J = -\frac{\tau_F}{\frac{d\omega}{dt}}$ and validated by the model.

Result Moment of inertia $J = 0.007 \text{ kgm}^2$

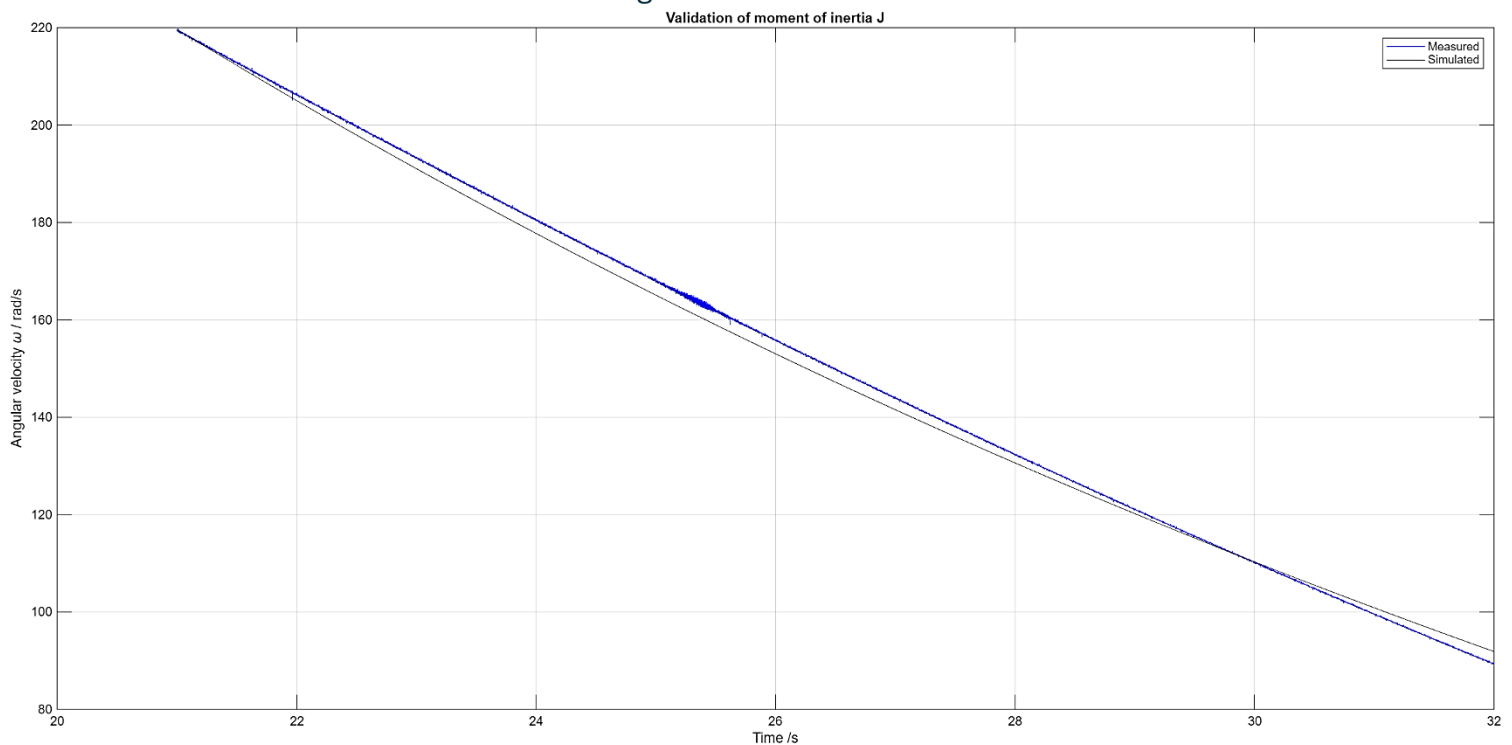


Figure 8: Validation of moment of inertia

Task 5: Simulation and Validation of Complete Model

The complete DC motor model was implemented in Simulink using the previously identified parameters R_A , L_A , k_G , τ_{F0} , α_F , J . The model was structured as a subsystem with inputs u_A and τ_L , and outputs i_A and n . The dry friction torque was implemented using sign and gain blocks to ensure correct direction. The model was simulated using the stepwise voltage excitation with step height of 3V and length of 3s. The simulated responses (2) were then compared with the measured data (1) for validation.

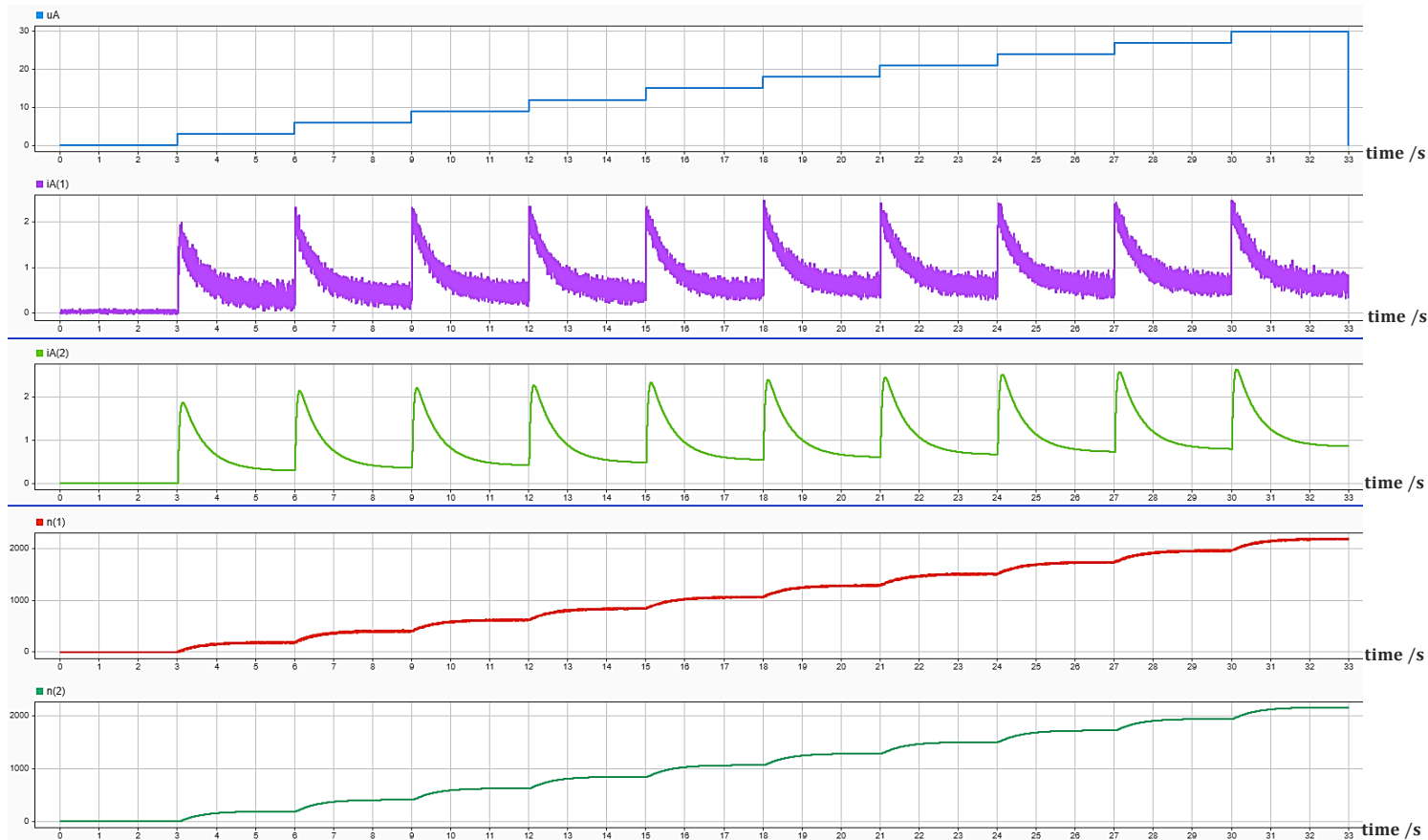


Figure 9: Measured (1) VS Simulated (2) responses

Observation and Remarks

Good agreement between simulated and measured speed response

Small deviations may be due to noise (e.g., quantization from ADC or sensor noise), method of parameter determination, low amount of data at operating points (angular velocity and armature current under no external load condition)

Current spikes in transients are better captured in the model than in real system due to physical limitation of power amplifier

Conclusion

The validated model successfully reproduces the dynamic behavior of the real DC motor, confirming that the identified parameters provide a physically consistent representation of the system.